

The Synthesis

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Independent

Abstract

What synthesis is not, and what it is.

A synthesis is not a proof.

A proof is a sequence of statements where each follows from the last. It ends. It has a conclusion. The conclusion is: this is true.

A synthesis is a map. It shows where the roads are. It shows where they stop. It does not build new roads. It does not claim to have crossed where no one has crossed. It says: here is what is known. Here is the exact boundary of what is known. The boundary is the wall.

This body is a synthesis.

The Millennium Synthesis: a synthesis of the open problems. Not their solution. Their anatomy.

Anatomy is valuable because the body you are studying is alive and you need to know where everything is before you can help it. The anatomy of an open problem is: where is the difficulty located? What are the symptoms? What would a cure look like?

We have answered these questions. For five of the six remaining Millennium Problems, the difficulty is located precisely. It has a name. For one of them (P

versus NP) the difficulty is not yet located. We know it is somewhere. We do not know where.

This is the synthesis.

A synthesis says: we have not been wasting time. The work that looks like it is circling the problem is not circling. It is building the map. And maps matter.

When the proof comes — if the proof comes — it will use this map. Not these specific papers. But the knowledge that the wall is here, not there. The knowledge of what the wall is made of. The knowledge of what has already been tried.

A synthesis is a gift to whoever comes next. It says: we looked. Here is what we found. Here is exactly where we stopped.

The rest is yours.